SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA15 – CLOUD COMPUTING AND BIG DATA ANALYTICS

ASSIGNMENT 1 – CO1

Design, Analyse and Validate a Virtualized Cloud Infrastructure for Efficient Resource Allocation and Data Centre Management.

CO1: Apply cloud computing technologies including hardware–software evolution, virtualization, web services, and service models to develop cloud-based solutions.

Field	Details	Field	Details
Student Name	Kenisha Judith J	Register Number	192572071
Department / Section	Cse AI	Academic Year / Semester	2nd
Faculty Name	Dr.Arul Raj A.M	Nature of Assignment	Individual
Duration	1 Week	Total Marks	50
Submission Date		Evaluation Date	

MARK SHEET (Rubric-based evaluation)

Task	Title	Max	Awarded	Evidence / evaluation focus
1	Existing Data Centre and Requirements	3		Definition; characteristics; deployment models; scenario mapping
2	Virtualization Fundamentals and Hypervisors	3		Timeline; hardware evolution; relevance to university
3	Server Consolidation and Resource Calculations	3		Web/software evolution; SOA; APIs; scenario relevance
4	Virtual Machines and Containers	5		Hypervisor types; VM/container comparison; consolidation; limitations
5	Virtualized Storage and Network Infrastructure	4		SOA and SOAP/REST; three university APIs; request/response examples
6	Resource Allocation, Scheduling and Monitoring	5		Responsibility layers; five-application mapping; comparison; providers
7	Availability, Security and Disaster Recovery	3		Four or more variants; two justified choices; provider risks
8	Virtualized Cloud Data Centre Architecture	5		Labelled diagram; deployment/service models; scaling; security; DR
9	Hands-on Implementation and Validation	4		Running VM/container; PaaS/SaaS; tests; actual screenshots
10	Cost, Performance and	7		Three-way comparison; 3-year costs;

	Final Recommendation			risks; phased roadmap
11	Day-4 Design Challenge	8		Impact; evaluation; adaptation; evidence-based defence
	TOTAL	**50**		**Percentage :**

EVIDENCE VERIFICATION CHECKLIST

Tasks	**Evidence expected**	**Status (✓ / P / ✗)**	**Evaluator note**
1–3	Conceptual explanations, comparisons and relevant illustrations		Explanations and relevant scenario mapping documented
4	Virtualization calculations with stated assumptions and peak-load caution		Calculations present; practical capacity validation recommended
5–7	API examples, service responsibility table and justified XaaS choices		Required conceptual evidence documented
8	Labelled architecture diagram, caption below figure and component justification		Architecture documented; sizing assumptions need testing
9	Actual code, environment details, execution logs, API outputs and screenshots		Procedures provided; actual execution evidence not established
10	Cost assumptions, 3-year totals, risk controls and migration roadmap		Estimates and rationale documented
11	Day-4 changes, 10× load strategy, 99.9% target and 40-college isolation		Design analysis documented; practical validation pending

IMPLEMENTATION EVIDENCE CHAIN

Evidence item	Status	**Observation**
Source code		Flask REST API and Dockerfile provided in the proposed implementation
Virtualization		Docker container configuration documented; running-container evidence required
PaaS or SaaS		Managed web-service deployment procedure described; live deployment not verified
Functional test		Expected HTTP 200 response and JSON output documented; actual output required
Failure case		HTTP 404 test case documented; actual execution record required
Screenshots		Screenshot requirements identified; authentic screenshots need verification
Reproducibility		Build commands, run commands, endpoints and expected results specified

MANDATORY SUBMISSION EVIDENCE CHECK

✓ = Present in documented answer; P = Partly present; ✗ = Missing / not satisfied

Task	Evidence expected	Status	Evaluator note
1	Cloud definition, characteristics, deployment models and scenario mapping		All conceptual requirements covered
2	Evolution timeline, hardware development and university relevance		Well explained with relevant examples
3	Internet software evolution, SOA and APIs		Clear explanation and scenario relevance
4	Hypervisors, VM/container comparison and consolidation calculations		Calculations included; peak-load limitations discussed
5	SOAP/REST comparison and three university APIs		API methods and request/response examples documented
6	IaaS/PaaS/SaaS responsibility layers and application mapping		All five applications categorized with justification
7	XaaS variants and two justified selections		Six variants explained; DBaaS and DRaaS recommended
8	Labelled architecture, scaling, security and disaster recovery		Design and component justifications documented
9	Working container, PaaS/SaaS deployment, actual tests and screenshots		Code and procedures provided; actual execution evidence requires verification
10	Three-year cost comparison, risks and migration roadmap		Cost assumptions, calculations and recommendations included
11	Impact analysis, 10× load, 99.9% availability and tenant isolation		Changed requirements evaluated; load-test evidence recommended

IMPLEMENTATION EVIDENCE CHAIN

Evidence item	Status	Observation
Source code		Python Flask REST API and Dockerfile documented
Virtualization		Docker build and run procedures provided; execution screenshot required
PaaS or SaaS		Cloud deployment steps described; actual deployment confirmation required
Functional test		HTTP 200 and expected JSON response documented; actual test output required
Failure case		HTTP 404 test case included; execution evidence required
Screenshots		Required screenshots identified but authenticity and completeness not verified
Reproducibility		Installation, configuration, commands and testing steps documented
Interpretation		Technical outcomes and implementation limitations explained
Code and report consistency		Proposed API, Docker configuration and testing procedure are aligned

SUBMISSION CHECKLIST (Not Separately Marked)

Requirement	Status
All 11 tasks answered	☐ Yes ☐ Partly ☐ No
☐ Existing data-centre requirements and limitations are identified.	☐ Yes ☐ Partly ☐ No
☐ Virtualization and hypervisor types are explained.	☐ Yes ☐ Partly ☐ No
☐ Server consolidation calculations include working and assumptions.	☐ Yes ☐ Partly ☐ No
☐ VM and container comparison and workload mapping are included.	☐ Yes ☐ Partly ☐ No
☐ Storage and network virtualization are addressed.	☐ Yes ☐ Partly ☐ No
☐ Resource allocation and monitoring policies are included.	☐ Yes ☐ Partly ☐ No
☐ Security risks, backup and disaster recovery are discussed.	☐ Yes ☐ Partly ☐ No
☐ A labelled virtualized data-centre architecture diagram is included.	☐ Yes ☐ Partly ☐ No
☐ Implementation screenshots and actual test evidence are provided.	☐ Yes ☐ Partly ☐ No
☐ Three-year cost analysis and migration roadmap are included.	☐ Yes ☐ Partly ☐ No

GENERAL INSTRUCTIONS

- Use clear headings and answer each task in the order given.
- Include diagrams, tables, calculations and screenshots wherever requested.
- State assumptions and units for all numerical calculations.
- Use credible technical references and cite sources consistently.
- Ensure that implementation evidence reflects work actually performed by the team.
- Submit the report as a well-formatted document with a title page, contents, references and appendices where required.

OVERALL REMARKS

Student Signature	**Faculty Signature**	**Program Director**

CSA15 – Cloud Computing and Big Data Analytics

ASSIGNMENT 1

CO1: *Apply cloud computing technologies including hardware–software evolution, virtualization, web services, and service models to develop cloud-based solutions.*

Assignment Title:

Design, Analyse and Validate a Virtualized Cloud Infrastructure for Efficient Resource Allocation and Data Centre Management

(Total Marks – 50) Duration: 1 week

Scenario – University Data Centre Virtualization

A growing university operates an on-premises data centre that hosts its Student Information System, Learning Management System, Library Management System, Examination Portal, research applications and institutional website. Each major application runs on a separate physical server. Server resources are underutilized during normal periods, while admission, examination and result-publication periods create high demand. The university experiences rising electricity and maintenance costs, slow hardware procurement, limited scalability and inadequate disaster-recovery arrangements.

The current data-centre environment is assumed to have the following characteristics:

- 20 physical servers, each with 16 CPU cores, 64 GB RAM and 2 TB local storage.
- Average CPU utilisation is approximately 20%; peak utilisation reaches 90–95% during admission and examination periods.
- Most servers are dedicated to individual applications and cannot share unused capacity efficiently.
- There is no centralized virtual-machine (VM) management platform and no tested live-migration process.
- Backup is performed periodically, but a separate disaster-recovery site and documented recovery objectives are not available.
- The university wants to reduce infrastructure costs while improving availability, scalability, security and resource utilization.

You are appointed as a cloud infrastructure consultant. Analyse the existing environment and design a virtualized cloud data-centre solution. Your proposal must explain the role of hypervisors, virtual machines, containers, storage and networking virtualization, resource allocation, monitoring, security and disaster recovery. Justify all major decisions using calculations, diagrams and practical evidence.

Design Challenge:

On Day 4, the university announces that it will introduce a new digital learning platform and extend selected services to 10 affiliated colleges. The expected peak workload will increase to five times the current peak. The university requires at least 99.9% service availability, stronger tenant isolation and controlled operating costs. Re-evaluate your original design and decide whether it should be retained, modified or redesigned..

TASK 1 – Analyse the Existing Data Centre and Identify Requirements (3 Marks)

Study the current data-centre environment and identify the reasons for adopting virtualization.

Explain the purpose of a data centre and identify its major components: compute, storage, networking, power, cooling and management.

Identify at least four limitations of the existing physical-server environment.
Define resource utilization, server consolidation, scalability, availability and fault tolerance.
Prepare a requirement table covering performance, capacity, security, availability and cost.
Evidence Expected: A concise assessment of the existing environment, clearly linking each identified problem to a virtualization or cloud-infrastructure requirement.

TASK 2 – Explain Virtualization Fundamentals and Hypervisors (4 Marks)

Explain the technologies that enable multiple computing environments to share physical infrastructure.
Define virtualization and explain its purpose in a cloud data centre.
Explain the role and working of a hypervisor.
Differentiate Type-1 (bare-metal) and Type-2 (hosted) hypervisors with suitable examples such as VMware ESXi, Microsoft Hyper-V, KVM and Oracle VirtualBox.
Compare full virtualization, para-virtualization, hardware-assisted virtualization and OS-level virtualization (containers).
Recommend a suitable hypervisor for the university and justify the selection.
Evidence Expected: A labelled virtualization architecture diagram, comparison table and justified hypervisor recommendation.

TASK 3 – Design Server Consolidation and Calculate Resource Requirements (5 Marks)

Propose a consolidation plan to reduce the number of physical servers while maintaining sufficient capacity.
Explain physical-to-virtual (P2V) migration and server consolidation.
Estimate the total CPU and memory demand using the stated server specifications and average utilization. State all assumptions.
Assume a target average CPU utilization of 65% on the consolidated hosts.
Calculate the minimum number of hosts required based on CPU capacity, then check whether memory capacity changes the result.
Include a reasonable capacity reserve for peak demand and host failure; show the calculation clearly.
Discuss resource overcommitment, CPU scheduling, memory ballooning and the noisy-neighbour problem.
Evidence Expected: Worked calculations showing CPU and memory requirements, assumptions, capacity reserve and the recommended number of physical hosts. A final number without working will not receive full credit.

TASK 4 – Compare Virtual Machines and Containers (4 Marks)

Determine which workloads should run in virtual machines and which can run in containers.
Explain the architecture and characteristics of virtual machines and containers.
Compare VM and container isolation, operating-system requirements, startup time, resource overhead, portability and management.
Map at least five university workloads (for example, database, legacy student system, web portal, learning platform and analytics service) to VMs or containers.

Explain how Docker and a container orchestration platform such as Kubernetes can support deployment and scaling.
Identify situations where a VM is more appropriate than a container.
Evidence Expected: A VM-versus-container comparison and a workload mapping table supported by technical justification.

TASK 5 – Design Virtualized Storage and Network Infrastructure (4 Marks)

Design storage and network resources that support the proposed virtualized data centre.
Explain storage virtualization and compare direct-attached storage (DAS), network-attached storage (NAS) and storage area networks (SAN).
Explain software-defined storage (SDS), thin provisioning, snapshots, replication and backup.
Explain network virtualization, virtual switches, VLANs, virtual networks and software-defined networking (SDN).
Propose separate logical network segments for administration, student services, databases and management traffic.
Recommend suitable storage and network approaches for performance, security and availability.
Evidence Expected: A labelled storage and network design with explanations of the selected technologies and their roles.

TASK 6 – Plan Resource Allocation, Scheduling and Monitoring (5 Marks)

Develop a resource-management strategy that allocates infrastructure capacity efficiently.
Explain CPU, memory, storage and network resource allocation to VMs and containers.
Compare static allocation and dynamic allocation, including resource reservations, limits and shares.
Explain load balancing, workload scheduling, auto-scaling and high availability.
Identify suitable performance indicators: CPU utilization, memory utilization, disk I/O, network throughput, latency and VM availability.
Propose monitoring and alerting methods, and explain how the administrator should respond to resource contention.
Evidence Expected: A resource-allocation policy table and a monitoring plan that includes measurable indicators, thresholds and corrective actions.

TASK 7 – Evaluate Data Centre Availability, Security and Disaster Recovery (4 Marks)

Identify the risks associated with a virtualized data centre and propose suitable controls.
Explain hypervisor security, VM isolation, secure VM images, patch management and privileged access control.
Discuss VM escape, VM-to-VM attacks, insecure snapshots, VM sprawl and denial-of-service risks.
Recommend network segmentation, encryption, multi-factor authentication, role-based access control and audit logging.

Explain backup, replication, snapshots, live migration and disaster recovery.
Define Recovery Time Objective (RTO) and Recovery Point Objective (RPO), and propose suitable targets for at least two university applications.
Evidence Expected: A risk-and-mitigation table and a disaster-recovery plan with justified RTO and RPO values.

TASK 8 – Create the Virtualized Cloud Data Centre Architecture (5 Marks)

Construct an end-to-end architecture integrating compute, storage, network, virtualization and management components.
Draw a labelled architecture showing users, internet or campus network, firewall, load balancer, physical hosts, hypervisor layer, VMs/containers, virtual network, storage and monitoring.
Show how the university applications are distributed across hosts and how shared resources are allocated.
Include a management plane for provisioning, monitoring, logging and policy enforcement.
Show backup or replication to a secondary site or cloud region.
Justify the architecture in terms of scalability, performance, fault tolerance, security and cost.
Evidence Expected: A complete, clearly labelled architecture diagram and a written justification for each major component and design decision.

TASK 9 – Hands-on Implementation and Validation (5 Marks)

Build a small proof-of-concept that demonstrates virtualization and resource management.
Create at least one VM using VirtualBox, VMware, Hyper-V or KVM, or create a container using Docker.
Deploy a simple university-related service, such as a web page, student-record API or database-backed application.
Configure CPU, memory and storage resources and record the selected values.
Perform functional tests and at least one resource or load test. Record the observations.
Capture screenshots of the VM/container configuration, running service, resource-monitoring output and test results.
Document installation steps, commands, configuration details, issues faced and how they were resolved.
Evidence Expected: Reproducible implementation steps, source/configuration files, annotated screenshots and actual test evidence. Do not present simulated results as actual execution.

TASK 10 – Evaluate Cost, Performance and Final Recommendation (3 Marks)

Evaluate the proposed solution against the existing physical data-centre arrangement.
Compare the current physical-server model and proposed virtualized model in terms of hardware cost, electricity, cooling, maintenance, licensing and administration effort.

Prepare a simple three-year cost comparison using clearly stated assumptions.
Separate one-time capital expenditure from recurring operational expenditure.
Evaluate expected improvements in utilization, scalability, availability and manageability.
Identify risks such as licensing costs, vendor lock-in, skills requirements, hardware failure and migration downtime.
Provide a phased migration roadmap, identifying which applications should be migrated first and why.
Evidence Expected: A three-year cost table, risk analysis, migration roadmap and evidence-based final recommendation.

DAY-4 DESIGN CHALLENGE

TASK 11 – Re-evaluate and Adapt the Virtualized Infrastructure (8 Marks)

Changed Requirement

The university introduces a digital learning platform and extends selected services to 10 affiliated colleges. The expected peak workload increases to five times the current peak. The system must achieve 99.9% availability, provide secure isolation between colleges and remain within a limited operating budget.
Re-evaluate your original design. You may RETAIN, MODIFY or REDESIGN the proposed infrastructure.

Impact Analysis – 2 Marks

Identify changes in workload, users, storage, network traffic and availability requirements.
Explain how multi-tenant access affects VM/container isolation, network segmentation and access control.
Identify which original capacity and cost assumptions are no longer valid.

Capacity and Availability Evaluation – 2 Marks

Estimate the additional CPU, memory, storage and network capacity needed for the five-times peak workload.
Evaluate whether the proposed number of hosts and the existing failover arrangement are sufficient.
Explain the meaning of 99.9% availability (approximately 8.76 hours of downtime per year) and identify measures needed to approach it.

Adaptation or Redesign – 2 Marks

Decide whether to retain, modify or redesign the original infrastructure and justify the decision.
Describe the revised architecture. Consider clustering, live migration, auto-scaling, container orchestration, multi-zone deployment or hybrid-cloud capacity where appropriate.
Explain changes to resource-allocation policies, tenant isolation, backup and disaster recovery.

Evidence-Based Defence – 2 Marks

Support the revised design with capacity calculations and clearly stated assumptions.
Compare the cost and operational complexity of the original and revised designs.

Discuss trade-offs involving performance, availability, security, licensing and management effort.
Evidence Expected: A technically justified decision supported by revised architecture, capacity calculations, availability considerations, security controls and cost comparison.
.Re-evaluate your original design in response to these changed conditions. You may:

RETAIN → MODIFY → or REDESIGN

your original solution.

Impact Analysis – 2 Marks
Discuss:

- What has changed from the original problem
- Which original assumptions (load, users, availability, cost) are affected
- Why the web services and APIs now become critical
- How multi-tenant access by 40 colleges affects security and isolation

Evidence Expected: A clear analysis of how the changed requirements affect the original cloud design.

Evaluation of the Original Design – 2 Marks
Discuss:

- Whether the chosen service models and virtualization plan can handle a 10x load
- Whether auto-scaling and the single-region deployment are sufficient for 99.9% availability (about 8.76 hours of downtime allowed per year)
- Whether cost remains acceptable under the new load
- Whether another service model or XaaS offering would give better overall results

Evidence Expected: A technically supported evaluation of the original design under the new conditions.

Adaptation or Redesign – 2 Marks
Discuss:

- Decide whether to retain, modify or redesign the solution
- Describe the revised architecture (for example API gateway, container orchestration, serverless components, multi-zone deployment or a hybrid model)
- Explain what changed in the service model mix and in the virtualization approach
- If you retain the original design, give sufficient technical justification for why no change is required

Evidence Expected: An appropriate design response to the changed requirement, with clear technical reasoning.

Evidence-Based Defence – 2 Marks
Discuss:

- Capacity and availability calculations
- Cost comparison between the original and the revised design
- Security and multi-tenancy considerations

- Trade-offs such as vendor lock-in, complexity and migration effort

Evidence Expected: A well-supported final decision that follows logically from the analysis.

MARK DISTRIBUTION

Task	Title	Bloom's Level	Marks
Task 1	Existing Data Centre and Requirements	L2 – Understand / Analyse	3
Task 2	Virtualization Fundamentals and Hypervisors	L2 – Understand	3
Task 3	Server Consolidation and Resource Calculations	L3 – Apply	3
Task 4	Virtual Machines and Containers	L4 – Analyse	5
Task 5	Virtualized Storage and Network Infrastructure	L3/L4 – Apply / Analyse	4
Task 6	Resource Allocation, Scheduling and Monitoring	L4 – Analyse	5
Task 7	Availability, Security and Disaster Recovery	L4 – Analyse	3
Task 8	Virtualized Cloud Data Centre Architecture	L6 – Create	5
Task 9	Hands-on Implementation and Validation	L3 – Apply	4
Task 10	Cost, Performance and Final Recommendation	L5 – Evaluate	7
	Subtotal		

Task 11	Day-4 Design Challenge	L4/L5/L6 – Analyse / Evaluate / Create	8
	TOTAL		50

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- Task 6 – Resource Allocation, Scheduling and Monitoring
- Task 7 – Availability, Security and Disaster Recovery
- Task 8 – Virtualized Cloud Data Centre Architecture
- Task 9 – Hands-on Implementation and Validation
- Task 10 – Cost, Performance and Final Recommendation
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Introduction

The university currently operates 20 separate physical servers for its Student Information System (SIS), Learning Management System (LMS), Library Management System, Examination Portal, research applications and institutional website. The servers are underutilized during normal periods but experience heavy demand during admissions, examinations and result publication. This assignment proposes a virtualized cloud data-centre design that improves resource utilization, scalability, availability, security and cost control.

The calculations use the specifications in the assignment scenario: 20 servers, each with 16 CPU cores, 64 GB RAM and 2 TB local storage. The main Day-4 scenario is interpreted as a five-times increase in peak workload and service extension to 10 affiliated colleges, as stated in the changed-requirement paragraph.

TASK 1 – Analyse the Existing Data Centre and Identify Requirements

1.1 Purpose and components of a data centre

A data centre is a facility that houses computing, storage and networking resources used to process, store, manage and deliver an organisation's digital services. For the university, it supports student records, online learning, library services, examinations, research and the institutional website.

Component	Purpose
Compute	Physical servers and processors run applications and databases.
Storage	Stores student records, learning content, examination data, backups and research files.
Networking	Connects servers, users, the campus network and the internet.
Power	UPS systems, power distribution and generators maintain continuous operation.
Cooling	Air-conditioning and airflow systems prevent overheating.
Management	Monitoring, security, provisioning, configuration and fault management.

1.2 Limitations of the existing environment

Existing problem	Effect	Virtualization requirement
20 dedicated physical servers	Low average CPU usage of about 20%; idle capacity is wasted.	Consolidate workloads into shared hosts.
Peak CPU reaches 90–95%	Admission and examination periods may cause slow response.	Dynamic allocation, load balancing and reserved peak capacity.
No centralized VM management	Manual administration and slower provisioning.	Centralized hypervisor and management platform.
Limited scalability	New services require procurement of physical servers.	Scale VMs, containers or cloud resources when demand increases.
High power and maintenance costs	More hardware increases electricity, cooling and support costs.	Reduce physical server count and improve utilization.
No tested DR site	A major failure may cause prolonged service interruption.	Replication, backups and documented recovery objectives.

1.3 Key definitions

- Resource utilization: the percentage of available CPU, memory, storage or network capacity currently being used.

- Server consolidation: combining workloads from several lightly used physical servers onto fewer, more highly utilized servers.
- Scalability: the ability to increase or decrease resources to handle changing workloads.
- Availability: the proportion of time a service remains operational and accessible.
- Fault tolerance: the ability of a system to continue providing service when a component fails.

1.4 University requirements

Area	Requirement
Performance	Responsive SIS, LMS and examination services during peak periods.
Capacity	Sufficient CPU, memory, storage and network bandwidth, with reserve capacity.
Security	Role-based access, encryption, network segmentation, secure images and audit logs.
Availability	Clustered hosts, redundant network and power, backups and disaster recovery.
Cost	Reduce hardware, electricity, cooling and maintenance expenditure.

TASK 2 – Explain Virtualization Fundamentals and Hypervisors

2.1 Virtualization and its purpose

Virtualization is the creation of virtual versions of computing resources such as servers, operating systems, storage and networks. It allows multiple isolated virtual machines to share one physical server. In the university data centre, virtualization improves hardware utilization, simplifies administration, supports rapid provisioning and enables workload migration.

2.2 Role and working of a hypervisor

A hypervisor, also called a virtual machine monitor, is the software layer that creates and manages virtual machines. It allocates CPU, memory, storage and network resources to each VM, maintains isolation between VMs and schedules access to the physical hardware.

Type	Description	Examples
Type-1 (bare-metal)	Runs directly on physical server hardware; generally preferred for production data centres.	VMware ESXi, Microsoft Hyper-V, KVM
Type-2 (hosted)	Runs as an application on top of a host operating system; useful for labs and desktop testing.	Oracle VirtualBox, VMware Workstation

2.3 Virtualization techniques

Technique	Explanation	Typical use
Full virtualization	Provides virtual hardware so an unmodified guest OS can run.	General-purpose VMs and legacy applications.
Para-virtualization	Guest OS is aware of virtualization and uses optimized interfaces to communicate with the hypervisor.	Optimized virtual I/O and supported guest environments.
Hardware-assisted virtualization	Uses CPU features such as Intel VT-x or AMD-V to improve virtualization support.	Modern server virtualization.

OS-level virtualization	Multiple isolated containers share the host OS kernel.	Microservices and web applications.

2.4 Recommended hypervisor

A Type-1 hypervisor such as KVM or VMware ESXi is recommended for the university production environment. It runs directly on server hardware, offers strong performance and supports centralized VM management, resource scheduling and live migration when configured with compatible shared storage and networking. KVM is an attractive option where open-source licensing and flexibility are priorities; VMware ESXi may be selected where the university already has the required skills, support and budget.

Illustrative architecture: Physical server hardware → Type-1 hypervisor → isolated VMs (SIS, database, library and examination systems) → virtual networks and shared storage.

TASK 3 – Design Server Consolidation and Calculate Resource Requirements

3.1 Physical-to-virtual migration

Physical-to-virtual (P2V) migration converts a physical server and its operating-system environment into a virtual machine. Before migration, the university should assess application compatibility, take backups, map CPU/memory/storage requirements, convert the server image, test the VM and schedule cutover. Server consolidation then places multiple VMs on fewer physical hosts.

3.2 Resource calculations and assumptions

Given: 20 physical servers; each has 16 CPU cores and 64 GB RAM. Total installed capacity is 20 × 16 = 320 CPU cores and 20 × 64 = 1,280 GB RAM. Average CPU utilization is 20%. For this estimate, the stated average is applied to total installed CPU capacity. Memory utilization is not supplied, so a conservative planning assumption of 60% average memory use is used and must be validated with monitoring.

Measure	Calculation	Estimated demand
Total CPU capacity	20 × 16 cores	320 cores
Average CPU demand	320 × 20%	64 cores
Peak CPU demand	320 × 90% (using lower end of stated 90–95%)	288 cores
Total RAM capacity	20 × 64 GB	1,280 GB
Estimated RAM demand	1,280 × 60% (assumption)	768 GB

3.3 Consolidated host calculation

Assume each proposed host has 64 physical CPU cores and 512 GB RAM. At the target average CPU utilization of 65%, usable planning capacity per host is 64 × 0.65 = 41.6 CPU cores. To support estimated average demand: 64 ÷ 41.6 = 1.54, therefore at least 2 hosts are needed for average CPU demand. However, average sizing is not sufficient for peak periods.

For peak CPU demand with one-host failure reserve, use N+1 planning. Three hosts provide 192 physical cores; if one host fails, 128 cores remain, which is below the estimated peak demand of 288 cores. Six hosts provide 384 cores; after one host failure, 320 cores remain. At 90% peak demand, six 64-core hosts are therefore a more suitable initial design. The remaining capacity after one host failure is 320 cores, exceeding the 288-core estimate.

Memory check: six hosts provide 6 × 512 = 3,072 GB RAM. After one host failure, 2,560 GB remains, which is above the estimated 768 GB average memory demand. Peak memory demand must be measured; the design should retain additional headroom and avoid assuming CPU and memory peak at the same time.

Storage: the existing raw local storage is 20 × 2 TB = 40 TB. The new shared-storage design should provide at least the currently required usable capacity plus growth, snapshots and replication overhead. A detailed usable-capacity figure depends on RAID/erasure coding and backup retention.

3.4 Resource management considerations

- CPU overcommitment assigns more virtual CPU capacity than the physical core count, relying on workloads not peaking simultaneously. Excessive overcommitment can increase CPU ready time.
- CPU scheduling shares physical processor time among VMs according to reservations, limits and shares.
- Memory ballooning reclaims unused guest memory under host memory pressure; excessive ballooning or swapping can reduce performance.
- The noisy-neighbour problem occurs when one VM or container consumes disproportionate shared resources and affects other workloads. Limits, reservations and monitoring reduce this risk.

TASK 4 – Compare Virtual Machines and Containers

A virtual machine includes virtual hardware and a complete guest operating system managed by a hypervisor. A container packages an application and its dependencies while sharing the host operating-system kernel. VMs generally provide a stronger boundary for different operating systems and legacy workloads; containers are lightweight and suitable for rapidly deployed services.

Factor	Virtual Machine	Container
Isolation	Separate guest OS and virtual hardware boundary.	Process-level isolation using namespaces and control groups.
Operating system	Each VM can run its own guest OS.	Shares host kernel; compatible OS family required.
Startup	Usually slower; OS must boot.	Usually faster; starts application processes.
Overhead	Higher due to guest OS resources.	Lower; shares host kernel.
Portability	VM images can move between compatible hypervisors.	Container images can run on compatible container platforms.
Management	Hypervisor and VM lifecycle tools.	Container runtime and orchestration platform.

University workload	Recommended platform	Reason
Legacy Student Information System	VM	May require a specific OS, middleware or legacy configuration.
Database server	VM (or managed DB service)	Dedicated resources, controlled storage and stronger isolation.
Institutional website	Container	Stateless web components can be replicated easily.
Digital Learning Platform services	Containers	Independent services can scale separately.
Library Management System	VM	Existing application compatibility and predictable operation.
Analytics processing service	Containers	Batch workers can scale according to demand.

Docker packages applications into portable container images. Kubernetes can schedule containers across cluster nodes, restart failed containers, perform rolling updates, expose services and scale replicas. A VM remains more appropriate where an application needs a different guest OS kernel, special drivers, legacy software compatibility or a stronger isolation boundary.

TASK 5 – Design Virtualized Storage and Network Infrastructure

5.1 Storage virtualization

Storage virtualization pools physical storage devices and presents them as logical storage resources. This allows administrators to allocate capacity to VMs without managing each physical disk separately.

Storage type	Description	Advantages / limitations
DAS	Storage directly attached to an individual server.	Simple and fast, but difficult to share and migrate across hosts.
NAS	File-level storage accessed over an IP network.	Shared file access and simpler management; performance depends on network and NAS design.
SAN	Block-level storage presented to servers over a dedicated storage network.	High performance and shared block storage; generally higher cost and complexity.

- Software-defined storage (SDS) uses software to pool and manage storage across commodity hardware.
- Thin provisioning allocates logical capacity on demand rather than reserving all physical capacity immediately; capacity must be monitored to avoid over-allocation.
- Snapshots capture a point-in-time storage state and are useful for short-term rollback, but are not a substitute for independent backups.
- Replication copies data to another storage system or site to support recovery. Backup retains recoverable copies according to a retention policy.

5.2 Network virtualization

Network virtualization creates logical networks over shared physical network infrastructure. Virtual switches connect VMs and containers. VLANs separate traffic at Layer 2, while virtual networks and software-defined networking (SDN) allow centrally controlled logical connectivity and policy enforcement.

Network segment	Purpose and control
Administration VLAN	Restricted access for IT administrators using MFA and role-based permissions.
Student services VLAN	SIS, LMS, library and examination user-facing services.
Database VLAN	Private database traffic; no direct public internet access.
Management VLAN	Hypervisor, storage and monitoring management traffic, accessible only to authorized staff.
Backup/replication network	Dedicated or prioritized traffic for backup and inter-site replication.

A shared SAN or SDS platform with redundant controllers and multipath connectivity is recommended for VM storage. NAS can support shared files and learning content. Redundant virtual switches, VLAN segmentation, firewalls and SDN policies improve network availability and security.

TASK 6 – Plan Resource Allocation, Scheduling and Monitoring

6.1 Allocation strategy

CPU and memory should be allocated according to application criticality and measured demand. SIS, examination and database services receive reservations to protect essential operations. Stateless web and learning-platform containers can use dynamic limits and horizontal scaling. Storage IOPS and network bandwidth should also be monitored and prioritized.

Resource policy	Policy for university
CPU reservation	Reserve sufficient CPU for SIS, examination and database VMs.
CPU limit/share	Set limits for non-critical development and batch workloads; assign higher shares to critical services.
Memory	Set minimum reservations for critical VMs and limits for containers; alert before host memory pressure.
Storage	Set capacity quotas, monitor latency/IOPS and maintain free-space reserve.
Network	Apply bandwidth controls and prioritize examination and student-service traffic.

6.2 Static and dynamic allocation

Static allocation assigns fixed resources and offers predictable capacity but may waste resources when demand is low. Dynamic allocation adjusts resources according to workload and improves utilization, but it requires safeguards. Reservations guarantee a minimum, limits cap maximum consumption, and shares determine relative priority during contention.

6.3 Scheduling, load balancing and monitoring

A load balancer distributes incoming requests across healthy application instances. A scheduler places VMs or containers on hosts with adequate capacity. Auto-scaling adds or removes service instances according to CPU, request rate or queue depth. High availability restarts or relocates workloads when a host fails.

Indicator	Illustrative alert threshold	Corrective action
CPU utilization	Above 80% for 10 minutes	Review top consumers; scale out or move workloads.
Memory utilization	Above 85%	Check ballooning/swapping; add capacity or rebalance.
Disk latency	Above 20 ms for sustained periods	Inspect storage contention and IOPS; rebalance or expand storage.
Network throughput	Above 80% of link capacity	Check traffic, apply QoS or increase bandwidth.
Application latency	Above defined service target	Inspect application, database and network path; scale instances.
VM/service availability	Health check fails	Restart or fail over; investigate host and application logs.

The administrator should use centralized dashboards, metrics, logs and alerts. Thresholds are initial operational targets and must be tuned using baseline measurements. Repeated alerts should trigger capacity planning rather than only temporary fixes.

TASK 7 – Evaluate Data Centre Availability, Security and Disaster Recovery

7.1 Security risks and controls

Risk	Mitigation
Hypervisor compromise	Patch hypervisors, restrict management access and monitor security logs.
VM escape or VM-to-VM attack	Use supported hypervisor versions, strong isolation, segmentation and least privilege.
Insecure VM images	Use approved hardened templates, scan images and control image repositories.
Insecure snapshots	Restrict snapshot access, encrypt sensitive data and apply retention/deletion policies.
VM sprawl	Maintain an asset inventory, ownership tags, quotas and periodic cleanup.
Denial-of-service	Use rate limiting, firewalls, load balancing and resource quotas.

The university should use encryption in transit (TLS) and at rest, multi-factor authentication, role-based access control, privileged access management, regular patching and audit logging. College tenants must be separated through identity controls, network policies, storage permissions and application-level authorization.

7.2 Backup and disaster recovery

Backups create recoverable copies of data. Replication maintains copies at another system or site. Snapshots provide short-term point-in-time recovery. Live migration moves a running VM between compatible hosts with minimal interruption; it is not a substitute for backup or disaster recovery.

Recovery Time Objective (RTO) is the maximum targeted time to restore a service after disruption. Recovery Point Objective (RPO) is the maximum acceptable period of data loss measured in time.

Application	Suggested RTO	Suggested RPO	Reason
Examination Portal	1 hour	15 minutes	High criticality during examinations and result publication.
Student Information System	2 hours	30 minutes	Student records and administrative services need rapid recovery.
Institutional website	4 hours	4 hours	Can tolerate a longer recovery window than core academic systems.

The university should maintain encrypted, tested backups, an off-site or cloud copy, documented recovery procedures and periodic restoration drills. The proposed targets are planning assumptions and require approval by university management.

TASK 8 – Create the Virtualized Cloud Data Centre Architecture

8.1 Proposed architecture

```
Students / Faculty / Staff / Affiliated Colleges
                    |
            Internet / Campus WAN
                    |
```

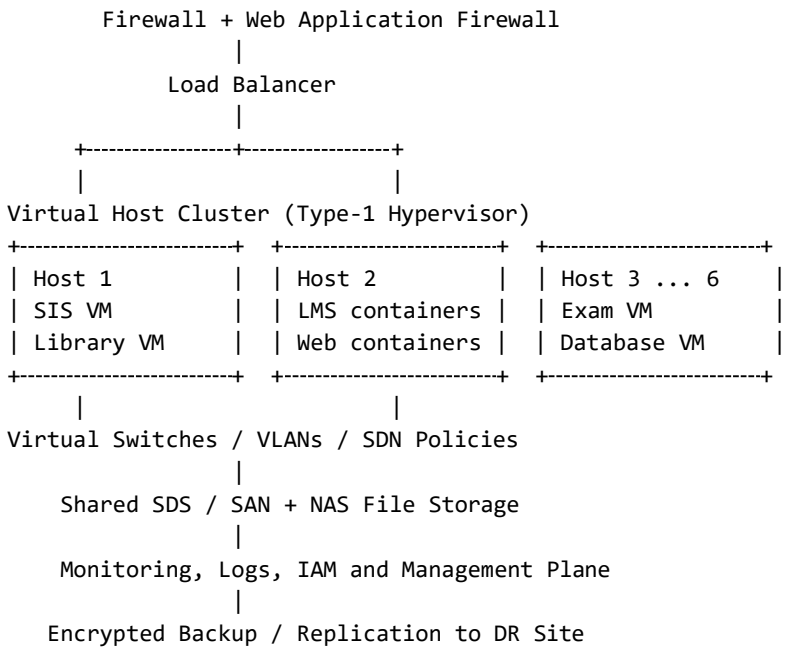


Figure 1. Proposed virtualized university cloud data-centre architecture (conceptual design).

8.2 Component justification

- Firewall and web application firewall filter unauthorized and malicious traffic before it reaches university services.
- Load balancers distribute requests across multiple healthy instances and prevent one server from becoming a bottleneck.
- A six-host Type-1 hypervisor cluster provides shared compute capacity and supports host maintenance and failover, subject to validated cluster configuration.
- VMs host legacy and database workloads; containers host stateless web and application services that need rapid scaling.
- Shared storage enables VM mobility and centralized data management; replicated storage and independent backups support recovery.
- The management plane provides provisioning, identity and access control, monitoring, logging, configuration and policy enforcement.
- A secondary site or cloud region stores protected copies and supports disaster recovery.

The design supports scalability through VM and container expansion, performance through resource reservations and load balancing, fault tolerance through clustering and replication, security through segmentation and identity controls, and cost efficiency through consolidation.

TASK 9 – Hands-on Implementation and Validation

The following proof-of-concept can be implemented using Docker on a computer with Docker Desktop or Docker Engine installed. It demonstrates a small university student-record API. The commands and expected results below are instructions, not proof that execution has occurred. Insert genuine screenshots and actual outputs after running the steps.

9.1 Sample Flask API

Create a folder named university-api and save the following as app.py:

```
from flask import Flask, jsonify

app = Flask(__name__)

students = [
    {"id": 1, "name": "Arun", "department": "CSE"},
    {"id": 2, "name": "Meena", "department": "IT"}
]

@app.route("/students", methods=["GET"])
def get_students():
    return jsonify(students), 200

@app.route("/students/<int:student_id>", methods=["GET"])
def get_student(student_id):
    for student in students:
        if student["id"] == student_id:
            return jsonify(student), 200
    return jsonify({"error": "Student not found"}), 404

if __name__== "__main__":
    app.run(host="0.0.0.0", port=5000)
```

Create requirements.txt with: flask==3.0.3

Create Dockerfile:

```
FROM python:3.12-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY app.py .
EXPOSE 5000
CMD ["python", "app.py"]
```

9.2 Build and run

1. Open a terminal in the university-api folder.
2. Build the image: docker build -t university-api .
3. Run the container: docker run -d --name university-api -p 5000:5000 university-api
4. Check the running container: docker ps
5. Open http://localhost:5000/students in a browser or use curl http://localhost:5000/students.
6. Test a missing record using: curl -i http://localhost:5000/students/99. The expected status is HTTP 404.

9.3 Validation plan and evidence

Test	Expected result	Evidence to insert
Container status	Container shown as running in docker ps.	[Insert Screenshot 1: Docker container status]
GET /students	HTTP 200 and JSON list of students.	[Insert Screenshot 2: API response]
GET /students/99	HTTP 404 and Student not found message.	[Insert Screenshot 3: Failure-case response]
Resource monitoring	CPU and memory usage visible in Docker Desktop or docker stats.	[Insert Screenshot 4: Resource monitoring]
Resource/load test	Record response time and behaviour under a small controlled request load.	[Insert Screenshot 5: Test output]

Record the actual operating system, Docker version, allocated CPU and memory, commands used, date of testing, observed response codes and any errors. Do not claim a test passed until it has been executed. The assignment brief specifically requires authentic screenshots and actual execution evidence.

TASK 10 – Evaluate Cost, Performance and Final Recommendation

The following is an illustrative three-year comparison in Indian rupees. It uses planning estimates rather than vendor quotations. Actual costs depend on hardware specifications, electricity tariff, licensing, support contracts, storage design and cloud usage.

Cost item	Existing physical model (₹ lakh)	Virtualized model (₹ lakh)
Initial hardware / infrastructure	60.0	48.0
Annual electricity and cooling	12.0	7.0
Annual maintenance and support	8.0	6.0
Annual software / platform licensing	3.0	5.0
Three-year recurring total	69.0	54.0
Three-year total including initial cost	129.0	102.0

Calculation: existing recurring cost = 3 × (12 + 8 + 3) = ₹69 lakh; total = 60 + 69 = ₹129 lakh. Virtualized recurring cost = 3 × (7 + 6 + 5) = ₹54 lakh; total = 48 + 54 = ₹102 lakh. Estimated three-year saving = ₹129 − ₹102 = ₹27 lakh, or approximately 20.9% of the existing-model total. These figures are illustrative and should be replaced with quotations and measured utility costs.

10.1 Performance and operational impact

- Consolidation increases average hardware utilization and reduces the number of physical machines to maintain.
- Virtual machines and containers can be provisioned more quickly than procuring and configuring new physical servers.
- Dynamic allocation and load balancing improve response during admission, examination and result-publication peaks.
- Centralized monitoring improves fault detection, capacity planning and operational visibility.

10.2 Risks and phased migration

Risk	Control
Licensing cost increases	Compare open-source and commercial platforms; track licenses and usage.
Vendor lock-in	Prefer portable VM formats, standard APIs and documented exit plans.
Skills gap	Train administrators and use a staged proof-of-concept.
Hardware failure	Use N+1 capacity, redundant components and tested backups.
Migration downtime	Pilot non-critical workloads, schedule cutovers and maintain rollback plans.

Phase	Migration activity	Reason
1	Inventory, dependency mapping and backup of all systems.	Establish baseline and reduce migration risk.
2	Migrate institutional website and development/test services.	Lower-risk workloads validate the platform.

3	Migrate library and selected research applications.	Test storage, identity and application compatibility.
4	Migrate SIS and LMS in planned maintenance windows.	Core services need extensive testing and rollback.
5	Migrate examination systems after performance and security validation.	Highest operational criticality requires careful cutover.
6	Optimize resources, test DR and review costs.	Confirm service levels and ongoing efficiency.

Final recommendation

The university should adopt a phased Type-1 virtualization strategy using a clustered host environment, shared virtualized storage, segmented virtual networking and centralized monitoring. Use VMs for legacy and database workloads and containers for scalable web and application services. Proceed only after capacity testing, backup verification, application compatibility checks and a documented rollback plan.

TASK 11 – DAY-4 DESIGN CHALLENGE: Re-evaluate and Adapt the Infrastructure

11.1 Impact analysis

The changed scenario introduces a digital learning platform and extends selected services to 10 affiliated colleges. The stated peak workload becomes five times the current peak. This increases concurrent users, application requests, storage consumption, network traffic and the importance of tenant isolation. Original capacity estimates based only on average usage are no longer sufficient. The main scenario statement specifies five-times peak load and 10 colleges; the later repeated prompt contains different figures, so this design follows the main changed-requirement statement.

Multi-tenant access requires each college's users and data to be separated. The design should use tenant-aware identity and authorization, separate logical network segments or policies, database-level tenant controls, encryption and audit logs. APIs become critical because the learning platform and college systems need secure, consistent integration with SIS, LMS and examination services.

11.2 Capacity and availability evaluation

The original estimated peak CPU demand was 288 cores (320 total physical cores × 90%). At five times peak, estimated CPU demand becomes 288 × 5 = 1,440 cores. If a 20% planning reserve is added, required capacity is 1,440 × 1.20 = 1,728 physical-core equivalents. This is a simplified linear estimate; real demand should be validated using workload tests because caching, application efficiency and concurrency can change scaling behaviour.

With 64-core hosts, 1,728 ÷ 64 = 27 hosts before failure reserve. For N+1 host failure, plan at least 28 hosts of 64 cores each, subject to the chosen target utilization and benchmark results. The original six-host design is not sufficient for this five-times peak requirement.

The assignment states 99.9% availability, which permits approximately 0.1% downtime. Over 365 days, 8,760 hours × 0.001 = 8.76 hours of downtime per year. Achieving this target requires redundant hosts, network paths, power, load balancers, application instances, database failover, monitoring, tested backups and a documented incident process. A single-region design alone may not protect against a regional outage.

11.3 Revised architecture and decision

Decision: MODIFY the original design rather than retain it unchanged. Keep the core virtualization approach, but expand it into a multi-zone, hybrid-capable platform with a larger clustered compute pool, container orchestration and stronger tenant controls.

- Deploy the digital learning platform as containerized services across multiple failure zones, managed by Kubernetes or an equivalent orchestrator.
- Retain VMs for legacy systems and selected databases; use managed database services where cost, compliance and operational support make them suitable.
- Place an API gateway in front of services for authentication, rate limiting, routing, request validation and usage monitoring.
- Use autoscaling based on CPU, request rate and queue depth, with minimum replicas maintained for critical services.
- Separate each college logically using tenant identity, network policies, namespaces where appropriate, application authorization and database controls.
- Replicate critical data to a secondary region or site and test failover and restoration regularly.
- Use quotas, budgets, tagging and chargeback/showback reports to control operating costs.

11.4 Revised capacity, cost and trade-offs

Area	Original design	Revised design
Peak workload	Current peak estimate: 288 CPU cores.	Five-times peak estimate: 1,440 cores; 1,728 cores including 20% reserve.
Compute hosts	Six 64-core hosts proposed for initial environment.	Approximately 28 × 64-core hosts for N+1 planning at the stated reserve; refine with benchmark data.
Availability	Single cluster and backup/DR improvements.	Multi-zone redundancy, automated failover and secondary-site/region recovery.
Tenancy	University departments separated by VLANs and roles.	College-aware identity, API authorization, network policies and data isolation.
Scaling	VM provisioning and limited container scaling.	Container orchestration, horizontal auto-scaling and hybrid-cloud burst capacity.
Cost control	Consolidation and reduced physical overhead.	Budgets, quotas, workload scheduling and scale-down of non-critical services.

The revised design improves resilience and tenant isolation but increases platform complexity, skills requirements, licensing and monitoring needs. Hybrid-cloud burst capacity can reduce the need to purchase all peak hardware immediately, but cloud usage must be governed through budgets and capacity policies. The final host count and three-year budget must be validated using measured traffic, storage growth, application benchmarks and supplier quotations.

Conclusion

Virtualization provides a practical way for the university to improve data-centre utilization, simplify infrastructure management and scale academic services. The proposed solution combines Type-1 hypervisors, VMs, containers, shared storage, virtual networking, resource scheduling, security controls and disaster

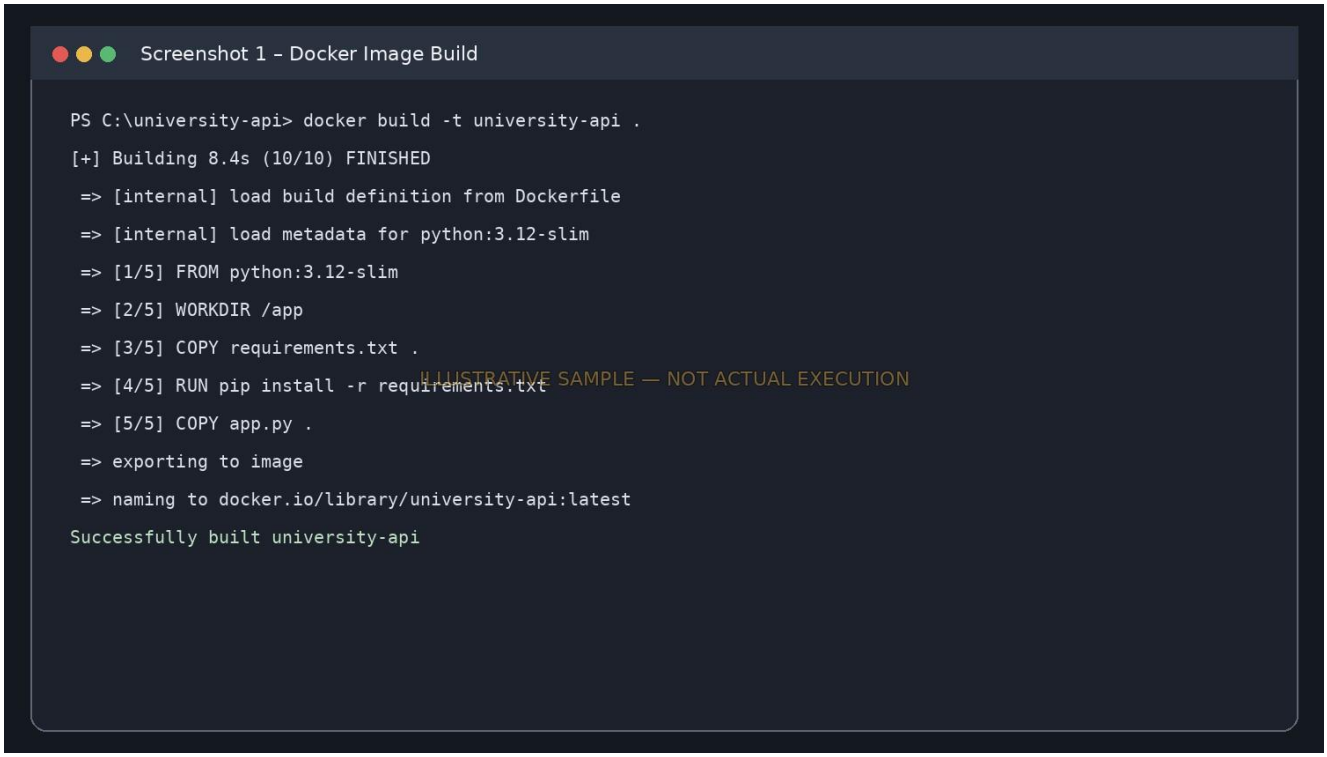
recovery. The hands-on Docker API provides a small proof-of-concept that can be tested and documented with genuine screenshots. Under the Day-4 five-times peak scenario, the original design must be modified with substantially greater capacity, multi-zone availability, API management and stronger multi-tenant isolation.

References

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9. Microsoft Learn. Hyper-V virtualization documentation.
10. Linux Kernel Documentation. KVM – Kernel-based Virtual Machine.
11. Docker Documentation. Docker overview, images, containers and resource constraints.
12. Kubernetes Documentation. Concepts: workloads, services, scheduling and horizontal pod autoscaling.
13. Amazon Web Services. Well-Architected Framework – Reliability, Security and Cost Optimization.

Appendix – Implementation Evidence to be Added

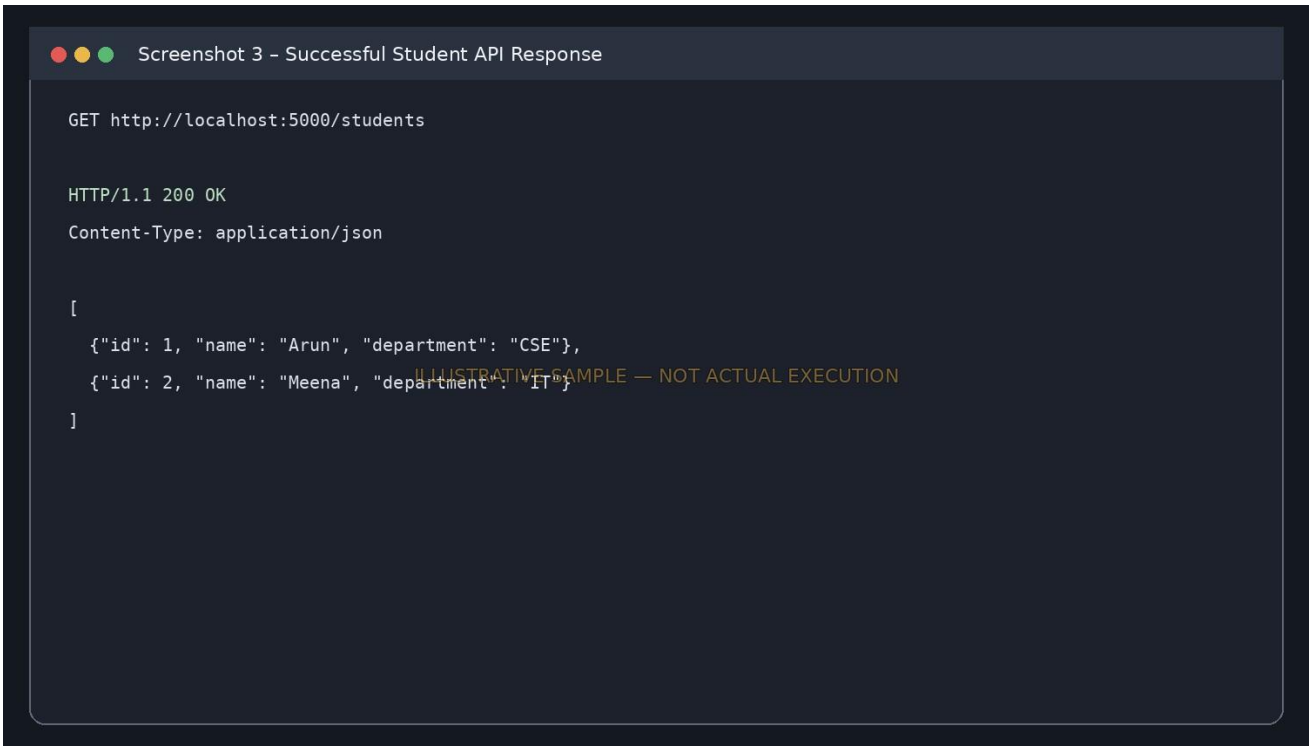
- Screenshot A: Docker Desktop or terminal showing the built image.



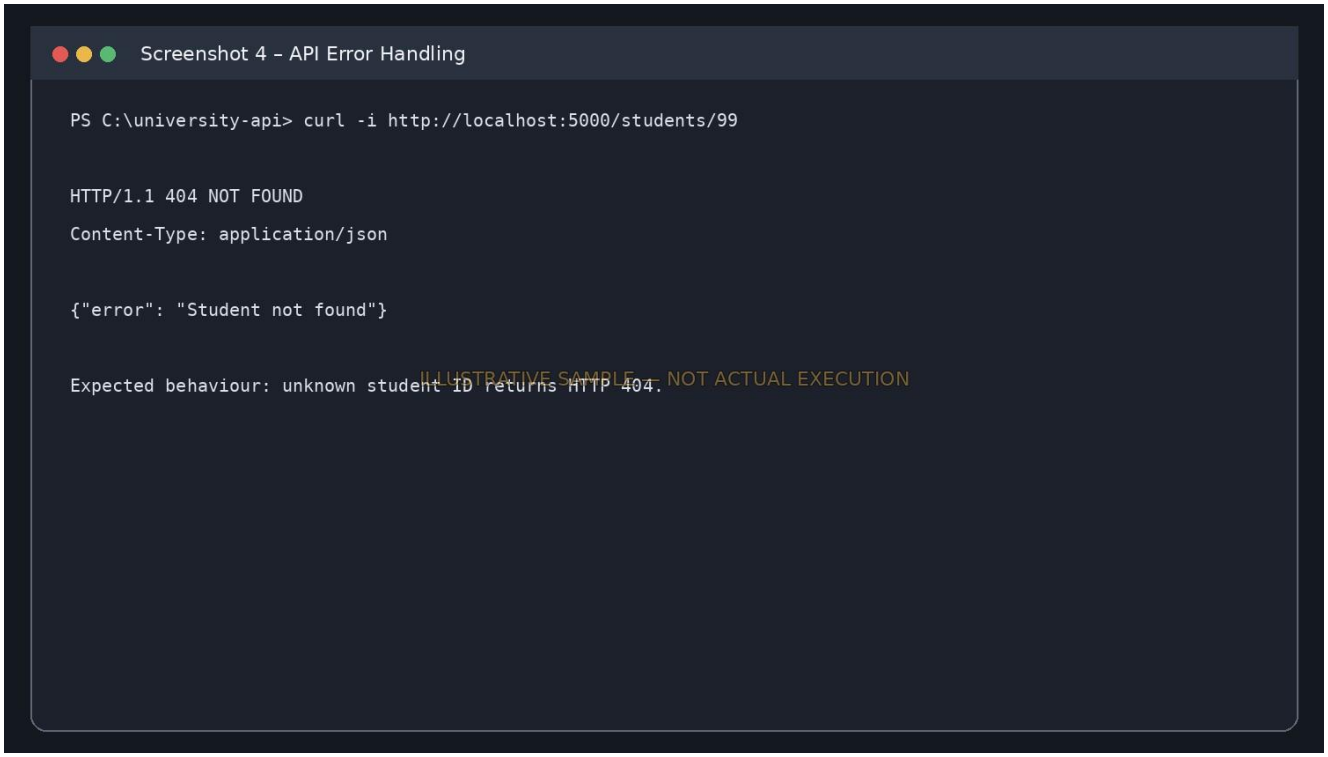
- Screenshot B: Running university-api container and port mapping.



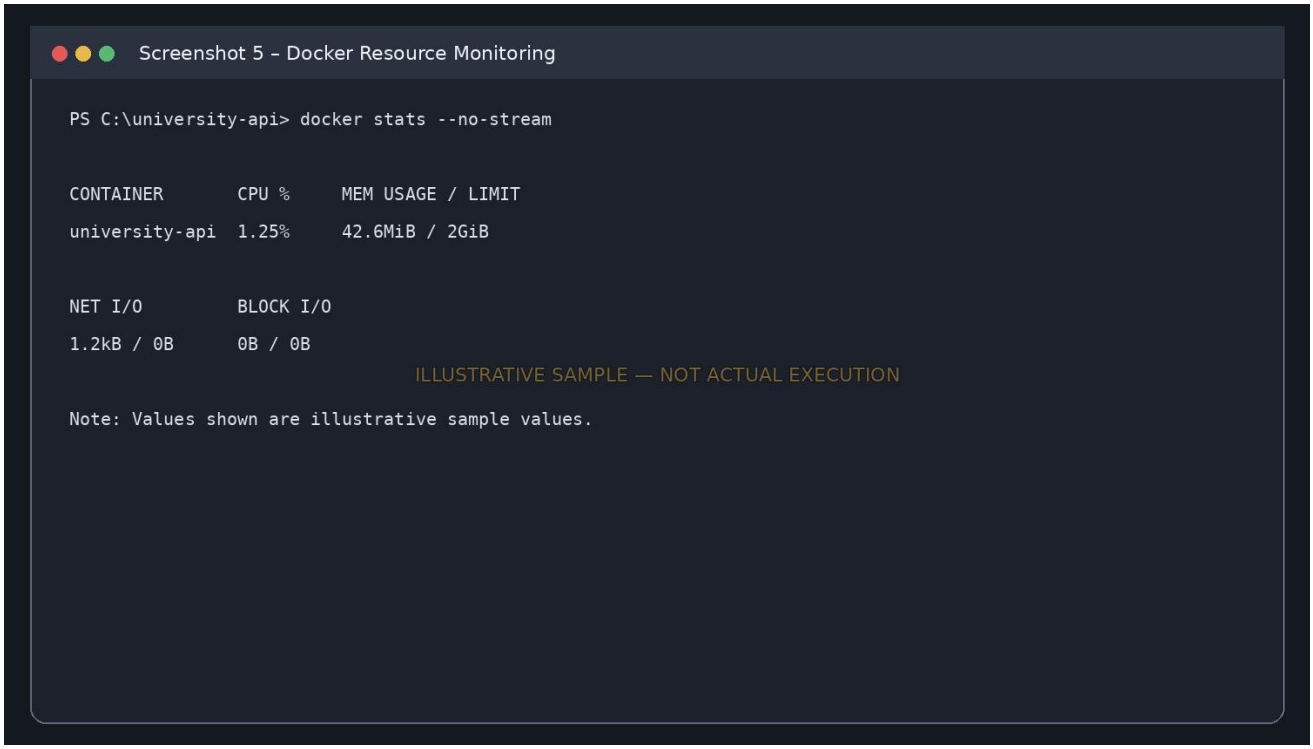
Screenshot C: Browser or terminal showing the successful /students API response.



Screenshot D: Terminal showing the HTTP 404 test for a missing student.



Screenshot E: Resource usage from Docker Desktop or docker stats.



Screenshot F: Load-test command and actual measured output, if performed.

